

Table 1. Device performance.

Material	Method	PCE (%)	Condition
MAPbI3	spin coating	21.3	AM 1.5G
FAPbI3	annealing	23.1	N2

Table 2. Stability summary.

Sample	Time (h)	Retention (%)
A	100	95
B	500	92
C	1000	88
D	2000	80

Table 3. Whitespace-only benchmark.

Model	RMSE	MAE
Baseline	0.15	0.12
Proposed	0.08	0.06

Table 4. Hyperparameters.

Param	Value	Unit	Range	Note
lr	1e-3	None	1e-4 to 1e-2	Adam
batch	32	None	16 to 128	fixed